Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOCHEMISTRY)
SEMESTER II EXAMINATION
BC 707: METABOLISM

Date: 30th April 2010
Time: 10.00 AM to 1.00 PM **Total Marks: 70**

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A **Total marks: 20**

Q1. Choose the correct option and put √ mark in front of it:

1) The coenzyme directly concerned with the synthesis of biogenic amines
a) TPP b) NADP+ c) Biotin d) Pyridoxal Phosphate

2) All of the following participate in the mitochondrial pyruvate dehydrogenase reaction except
a) Flavin b) Coenzyme A c) Lipoamide d) Biotin

3) Which of the following participate in one carbon transfer reaction?
a) Tetrahydrofolate b) Lipoamide c) Falvin d) TPP

4) In TCA cycle, which is first formed
a) Isocitrate b) Citrate c) Succinate d) Fumarate

5) Allosteric inhibitors of phosphofructokinase-1 includes all except
a) Citrate b) Fatty acid c) AMP d) Ketone bodies

6) Glycogen synthetase requires as substrate
a) UDP-glucose b) GDP-glucose c) ADP-glucose d) CDP-Glucose

7) Which of the following enzymes function in gluconeogenesis forms glucose phosphate
a) Aldolase b) Phosphoglucomutase
c) Triose-phosphate dehydrogenase d) All of the above

8) Which of the following compounds is not produced from dietary starch by salivary alpha-amylase?
a) Maltose b) Isomaltose c) Malto triose d) Fructose

9) Net gain of ATP in β oxidation of Palmitic acid is
a) 35 b) 96 c) 131 d) 129

10) Rate limiting step of fatty acid synthesis is mediated by
a) Enoyl Reductase b) Acetyl CoA carboxylase
c) Thioesterase d) Ketoacyl reductase

11) Number of NADPH consumed in synthesis of one molecule of Palmitate form acetyl CoA and Malonyl CoA is
a) 32 b) 8 c) 14 d) 2

12) Ketogenesis occurs in
a) Liver only b) Adipose Tissue only
c) Skeletal Muscle only d) All of the above

13) Catabolism of following amino acid forms acetyl CoA except
a) Tyrosine b) Tyrptophan c) Lysine d) Cysteine

14) Maple syrup urine disease is a deficiency of which amino acid degradative pathway?
a) tyrosine b) leucine c) tryptophan d) arginine

15) Which of the following amino acids is likely responsible for most transamination reactions in a cell?
a) glutamate b) glutamine c) aspartate d) alanine

16) In the synthesis of cysteine, the carbon skeleton is provided by
a) Serine b) Methionine c) Glutamate d) Alanine

17) The end product of purine metabolism in humans is
a) Xanthine b) Uric acid c) Urea d) Allantoin

18) Adenylosuccinate synthetase requires
a) ATP b) GTP c) CTP d) TTP

19) The nitrogen atoms in the purine ring are obtained from
a) Glycine b) Glutamine c) Aspartate d) All of the above

20) Which of the following bases cannot be directly salvaged?
a) guanine b) hypoxanthine c) xanthine d) adenine

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Part B **Total Marks: 50**

Section I

Q1 A Answer Any TWO:

a) Give structure of Carnitine. Explain its role in transport of the fatty acid into mitochondria. [4]

b) Briefly describe synthesis of ketonebodies. [4]

c) Give an outline of β – oxidation of Arachidic acid [4]

B Write short note on Prostaglandins [2]

Q2 A Answer any Two:

a) Role of ribonucleotide reductase [3]

b) Give overview of integration of metabolic pathways of energetic metabolism [3]

c) Regulation of purine nucleotide biosynthesis [3]

B Discuss degradation of Purine nucleotides [4]

Section II

Q3 A Write short note on:

a) Pyridoxal phosphate-A Coenzyme [5]

OR

b) Tracing techniques for studying metabolism [5]

B How Thiamine pyrophosphate involved in Oxidative decarboxylation reaction? Explain with suitable example. [5]

Q4 A Answer Any Two:

a) List distinctive features of glucokinase and hexokinase [4]

b) Discuss how PFK 1 activity is regulated [4]

c) Write short note on Glyoxylate pathway [4]

B Write following reaction of TCA cycle [2]

- Oxidative decarboxylation of alpha ketoglutarate
- Oxidation of Succinate

Q5 A Explain biochemical basis of Ammonia Toxicity [2]

B Write short notes on: (Any Two)

Mechanism of Transamination reaction [4]

Glutamine synthetase [4]

Biosynthesis of Phynylalanine and tyrosine from chorismate [4]